



Project Documentation

Predictive Workforce Transition: AI-Powered Job Displacement & Career Reskilling

10Alytics x JengaGlobal Hackathon 2026 — Track C (Data Science)

Client Scenario: StrataWork Global

1 The Problem

StrataWork Global automated roughly 65% of entry-level analytics tasks after adopting generative AI, improving turnaround by 300% but leaving hundreds of junior and mid-level analysts facing displacement with no structured path forward. Leadership has no empirical way to identify which roles are most exposed, or which adjacent roles offer a realistic transition.

Our Solution: A system that (1) classifies occupational automation risk as High, Medium, or Low, (2) maps displaced workers to realistic adjacent career paths based on their actual skills, and (3) gives talent leaders a company-wide view of exposure all through one interactive application.

2 Approach

We followed the five-stage workflow set out in the case study brief: data ingestion, model training, skill mapping, interface build, and validation. At every stage, we prioritized honest validation over flattering numbers; a decision that shaped the entire project and is detailed in Section 3.

3 Exploratory Data Analysis & Feature Engineering

The dataset (`ai_job_market_insights.csv`, 500 records, 10 fields) was clean: zero missing values, zero duplicate rows, no inconsistent category spellings. No data cleaning was required.

Critical Finding: Before engineering any features, we tested whether each column actually relates to the target (`Automation_Risk`) using chi-square tests (categorical features) and ANOVA (numeric features). The result: almost none of the provided columns including `AI_Adoption_Level1`, the column most intuitively expected to matter showed a statistically meaningful relationship with the target. Only `Job_Title` and `Salary_USD` passed the standard significance threshold, and only marginally.

Engineered Feature Testing: An earlier version of this project engineered a composite `AI_Pressure_Index` feature (a weighted combination of AI adoption level and job growth projection). Direct testing showed this feature carried no signal either (High/Medium/Low risk groups averaged 0.582 / 0.593 / 0.585 — statistically indistinguishable) — expected, since it was built from two inputs that were individually uninformative. It was removed rather than retained for appearance.

External Data Enrichment: Per guidance from our hackathon coach that sourcing outside data was both permitted and encouraged, we built `job_risk_reference.py`, mapping each job title to a Low/Medium/High automation-risk tier grounded in Frey & Osborne's published occupational automation research, with each mapping's basis documented (direct citation where a match exists, reasoned judgment where the modern job title has no 2013-era equivalent). Tested the same honest way: it showed no significant relationship with the dataset's `Automation_Risk` label ($p=0.076$) and did not improve model accuracy when included (35.25% vs. 36.00% baseline) expected, since it is fully derived from `Job_Title`, a feature the model already uses. Rather than discard this work, it is surfaced in the application as an independent, transparently-sourced second opinion alongside the model's own prediction.

4 Trained ML Classification Model

Approach: Logistic Regression and Random Forest classifiers were trained on the verified feature set (Salary_USD, Job_Title, Industry, Company_Size, Location, AI_Adoption_Level, Required_Skills, Remote_Friendly, Job_Growth_Projection), evaluated using 5-fold stratified cross-validation against a majority-class baseline not a single train/test split, which can overstate performance by chance.

Model	5-Fold CV Accuracy
Majority-class baseline	34.50% ($\pm 0.61\%$)
Logistic Regression (L2)	36.00% ($\pm 4.96\%$)
Random Forest	34.75% ($\pm 5.02\%$)

Honest Interpretation: Logistic Regression's advantage over the baseline is small and falls within one standard deviation of it a genuine but limited edge, not a highly accurate predictive model. A single held-out test split showed 44% accuracy; we report the cross-validated figure as the reliable one, since it does not depend on one favorable split. Holdout confusion matrix and per-class metrics are included in the Phase 2 notebook for full transparency.

5 Skill Adjacency & Career Mapping System

Implemented in `skill_engine.py`. Because the dataset records only one skill per job (not a full skill profile), genuine statistical skill-overlap computation was not possible from the data alone. Role competency profiles were instead curated from standard occupational skill sets, with this design choice disclosed rather than presented as data-derived.

The engine is **risk-aware**: it receives the model's predicted risk level and routes recommendations accordingly High/Medium risk routes toward automation-resilient pivot roles; Low risk routes toward vertical, specialization-focused growth roles. Matching skills and skill gaps are computed and surfaced for each recommended role.

6 Interactive Application

Built in Streamlit (`app.py`), with three views:

1. Workforce Mobility Portal

Individual risk assessment and risk-aware career pathway recommendations, including the external research reference described in Section 3.

2. Talent Analytics Dashboard

Company-wide exposure distribution by risk tier and by industry, for talent leaders.

3. Methodology & Benchmarks

Full, honest disclosure of model performance, feature selection decisions, and the external reference feature's tested (negative) result on model accuracy.

7 Workforce Insights & Visualizations

- **Workforce distribution by exposure band:** Risk tier counts across the organization.
- **Exposure by industry vertical:** Comparative vulnerability analysis across sectors.
- **Feature-vs-target relationship charts:** EDA documentation establishing which factors do and do not relate to displacement risk.

8 Technical Documentation Summary

- **Preprocessing:** Standard scaling for numeric features, one-hot encoding for categorical features, via a single `ColumnTransformer` pipeline.
- **Feature engineering:** Limited to verified, tested features; two candidate engineered features (`AI_Pressure_Index`, `Salary_Tier`) were built, tested, and explicitly discarded after failing to show signal.
- **Model selection:** Compared via 5-fold stratified cross-validation against an explicit baseline, not single-split accuracy.
- **Evaluation:** Accuracy, macro F1, confusion matrix, and per-class precision/recall.
- **Recommendation logic:** Rule-based, risk-aware skill matching (Section 5).
- **Deployment:** Streamlit Community Cloud, deployed from GitHub.

9 Disclosure of AI Tool Use

In line with the hackathon rule that AI tool use is permitted but must be disclosed:

Google Gemini

Draft Generation

Used to generate initial drafts of the EDA notebook, model training notebook, and skill-matching logic.

Claude (Anthropic)

Auditing & Refinement

Used extensively throughout to: audit the Gemini-generated code for statistical rigor and bugs (including identifying that the original approach lacked any feature-target relationship testing, and catching a notebook bug where stale code would have silently overwritten a corrected model file); conduct the baseline comparison and cross-validation analysis; source and integrate the external research reference; rewrite the application's visual design; and draft this documentation.

Google Antigravity

Design Iteration

Used for one design-refinement attempt on the application's visual styling; this attempt introduced a text-rendering defect and was discarded in favor of the previously verified design.

All modeling decisions, statistical tests, and their honest interpretation were reviewed and verified by the team before inclusion.

10 Expected Business Value

This system allows StrataWork Global to move from **reactive restructuring to proactive workforce planning**: flagging likely at-risk roles early, giving displaced or at-risk employees a concrete, skills-based transition path rather than no path at all, and giving talent leaders company-wide visibility into exposure while being transparent about the current model's real, limited predictive power rather than overstating it.